QUANTUM COMPUTING

AN INTRODUCTION

Quantum Fourier Transform

RAFAEL HERNÁNDEZ HEREDERO, RAFAEL DELGADO LÓPEZ
ETSIS de Telecomunicación, UPM

April 25, 2024

Outline

Outline

Outline

Outline

Definition of DFT

Discrete Fourier Transform

- Periodic signal (sound, electromagnetism, electronics): useful for conversion between time (Amplitude vs. Time) and frequency (Amplitudes vs. Frequency) spaces.
- Partial Differential Equations: turns differential equations into algebraic equations. Specially useful for periodic boundary conditions.
- Quantum Mechanics: conversion between position and momentum spaces.

Definition of DFT

Discrete Fourier Transform

- Periodic signal (sound, electromagnetism, electronics): useful for conversion between time (Amplitude vs. Time) and frequency (Amplitudes vs. Frequency) spaces.
- Partial Differential Equations: turns differential equations into algebraic equations. Specially useful for periodic boundary conditions.
- Quantum Mechanics: conversion between position and momentum spaces.

Classical implementation

$$\phi_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} a_j e^{2\pi i \cdot jk/N}$$

Definition of DFT

Inverse DFT

- The Discrete Fourier Transform is invertible.
- If ϕ_k (for $k = 0, 1, 2, \ldots, N-1$) is the DFT of a_j ($j = 0, 1, 2, \ldots, N-1$), then the original a_j can be recovered via the Inverse DFT (IDFT):

Definition of DFT

Inverse DFT

- The Discrete Fourier Transform is invertible.
- If ϕ_k (for $k = 0, 1, 2, \ldots, N-1$) is the DFT of a_j ($j = 0, 1, 2, \ldots, N-1$), then the original a_j can be recovered via the Inverse DFT (IDFT):

Inverse Discrete Fourier Transform

$$a_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \phi_k e^{-2\pi i \cdot kj/N}$$

Definition of DFT

Inverse DFT

- The Discrete Fourier Transform is invertible.
- If ϕ_k (for $k = 0, 1, 2, \ldots, N-1$) is the DFT of a_j ($j = 0, 1, 2, \ldots, N-1$), then the original a_j can be recovered via the Inverse DFT (IDFT):

Inverse Discrete Fourier Transform

$$a_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \phi_k e^{-2\pi i \cdot kj/N}$$

Proof: Intermediate Step

$$\sum_{l=0}^{N-1} \exp(2\pi i \cdot kl) = \begin{cases} N, & \text{iff } k = qN, q \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}$$

Properties of the DFT

DFT as a linear transformation

- The DFT can be interpreted as a linear transformation, where $\boldsymbol{a} = (a, a_1, \ldots, a_{N-1})$ and $\boldsymbol{\phi} = (\phi_0, \phi_1, \ldots, \phi_{N-1})$

Properties of the DFT

DFT as a linear transformation

- The DFT can be interpreted as a linear transformation, where $\boldsymbol{a} = (a, a_1, \ldots, a_{N-1})$ and $\boldsymbol{\phi} = (\phi_0, \phi_1, \ldots, \phi_{N-1})$
- The linear transformation is unitary.

Properties of the DFT

DFT as a linear transformation

- The DFT can be interpreted as a linear transformation, where $\boldsymbol{a} = (a, a_1, \ldots, a_{N-1})$ and $\boldsymbol{\phi} = (\phi_0, \phi_1, \ldots, \phi_{N-1})$
- The linear transformation is unitary.
- In this space, and with the previous definition of DFT, the canonical matrix of the DFT is:

Properties of the DFT

DFT as a linear transformation

- The DFT can be interpreted as a linear transformation, where $\boldsymbol{a} = (a, a_1, \ldots, a_{N-1})$ and $\boldsymbol{\phi} = (\phi_0, \phi_1, \ldots, \phi_{N-1})$
- The linear transformation is unitary.
- In this space, and with the previous definition of DFT, the canonical matrix of the DFT is:

Properties of the DFT

DFT as a linear transformation

- The DFT can be interpreted as a linear transformation, where $\boldsymbol{a} = (a, a_1, \ldots, a_{N-1})$ and $\boldsymbol{\phi} = (\phi_0, \phi_1, \ldots, \phi_{N-1})$
- The linear transformation is unitary.
- In this space, and with the previous definition of DFT, the canonical matrix of the DFT is:

Canonical Matrix of DFT

If $\omega = e^{2\pi i/N}$, then

$$\mathrm{(DFT)_{kj}} = \frac{1}{\sqrt{\mathrm{N}}}\omega^{\mathrm{kj}}$$

$$\mathrm{(IDFT)_{kj}} = \frac{1}{\sqrt{\mathrm{N}}}\omega^{-\mathrm{kj}}$$

Properties of the DFT

Canonical Matrix of DFT

If $\omega = e^{2\pi i/N}$, then

$$(\mathrm{DFT})_{\mathrm{kj}} = \frac{1}{\sqrt{\mathrm{N}}} \begin{pmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega^1 & \omega^2 & \omega^3 & \dots & \omega^{(N-1)} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \dots & \omega^{2(N-1)} \\ 1 & \omega^3 & \omega^6 & \omega^9 & \dots & \omega^{3(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{(N-1)} & \omega^{2(N-1)} & \omega^{3(N-1)} & \dots & \omega^{(N-1)(N-1)} \end{pmatrix}$$

Properties of the DFT

Example, $N = 4$, DFT

$$\mathrm{DFT} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

$$\mathrm{IDFT} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix} = (\mathrm{DFT})^\dagger$$

Outline

Quantum Fourier Transform

Operation on a quantum state

- Consider a qubit register of n qubits initialized to state:

$$|\mathbf{\Psi}\rangle = a_0 |0\rangle + a_1 |1\rangle + \cdots + a_{N-1} |N-1\rangle ,$$

where $N = 2^n$. Note that this requires the normalization of the $\boldsymbol{a}$ vector.

Quantum Fourier Transform

Operation on a quantum state

- Consider a qubit register of n qubits initialized to state:

$$|\mathbf{\Psi}\rangle = a_0 |0\rangle + a_1 |1\rangle + \cdots + a_{N-1} |N-1\rangle \,,$$

  where $N = 2^n$. Note that this requires the normalization of the $\boldsymbol{a}$ vector.

- The QFT operator transforms this register to state

$$|\boldsymbol{\phi}\rangle = \phi_0 |0\rangle + \phi_1 |1\rangle + \cdots + \phi_{N-1} |N-1\rangle$$

Quantum Fourier Transform

Operation on a quantum state

- Consider a qubit register of n qubits initialized to state:

$$|\Psi\rangle = a_0 |0\rangle + a_1 |1\rangle + \cdots + a_{N-1} |N-1\rangle ,$$

  where $N = 2^n$. Note that this requires the normalization of the $\boldsymbol{a}$ vector.

- The QFT operator transforms this register to state

$$|\phi\rangle = \phi_0 |0\rangle + \phi_1 |1\rangle + \cdots + \phi_{N-1} |N-1\rangle$$

- This transformation is unitary because DFT itself is unitary.

Quantum Fourier Transform

Operation on a quantum state

- Consider a qubit register of n qubits initialized to state:

$$|\mathbf{\Psi}\rangle = a_0 |0\rangle + a_1 |1\rangle + \cdots + a_{N-1} |N-1\rangle,$$

  where $N = 2^n$. Note that this requires the normalization of the $\boldsymbol{a}$ vector.

- The QFT operator transforms this register to state

$$|\boldsymbol{\phi}\rangle = \phi_0 |0\rangle + \phi_1 |1\rangle + \cdots + \phi_{N-1} |N-1\rangle$$

- This transformation is unitary because DFT itself is unitary.
- The IQFT can be implemented as the conjugate transpose of the QFT operator.

Quantum Fourier Transform

Operation on a quantum state

- Consider a qubit register of n qubits initialized to state:

$$|\Psi\rangle = a_0 |0\rangle + a_1 |1\rangle + \cdots + a_{N-1} |N-1\rangle ,$$

  where $N = 2^n$. Note that this requires the normalization of the $\boldsymbol{a}$ vector.

- The QFT operator transforms this register to state

$$|\phi\rangle = \phi_0 |0\rangle + \phi_1 |1\rangle + \cdots + \phi_{N-1} |N-1\rangle$$

- This transformation is unitary because DFT itself is unitary.
- The IQFT can be implemented as the conjugate transpose of the QFT operator.
- However, **be cannot recover the DFT transformed vector** ϕ, at least in a direct way and with a single measurement.

Quantum Fourier Transform

Operation on a quantum state

If $|a\rangle = \sum_{j=0}^{N-1} a_j |j\rangle$, then the QFT action over each vector $|k\rangle$ is

$$\mathrm{QFT}\,|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i \cdot jk/N} |k\rangle$$

The usual manipulations for recovering the QFT involve:

$$j = (j_{n-1}j_{n-2}\dots j_1 j_0)_2 = \sum_{l=0}^{n-1} j_l \cdot 2^l, \qquad k = \sum_{l=0}^{n-1} k_l \cdot 2^l$$

$$\frac{j}{2^n} = \sum_{l=0}^{n-1} j_l \cdot 2^{-(n-l)} = (0.j_{n-1}\dots j_1 j_0)_2, \qquad e^{2\pi i \cdot z} = 1\, \forall z \in \mathbb{Z}$$

Then

$$\mathrm{QFT}\,|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i (0.j_0) k_{n-1}} \cdot e^{2\pi i (0.j_1 j_0) k_{n-2}} \dots e^{2\pi i (0.j_{n-1}j_{n-2}\dots j_1 j_0) k_0} |k\rangle$$

QFT implementation

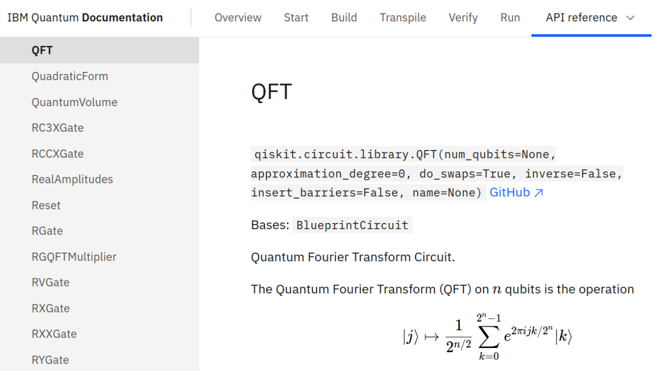
QFT implementations on the bibliography

- Very detailed description: [Thomas C. Wong]

QFT implementation

QFT implementations on the bibliography

- Very detailed description: [Thomas C. Wong]
- QUIRK and QISKIT: QFT and IQFT are standard blocks.


IBM Quantum **Documentation** | Overview | Start | Build | Transpile | Verify | Run | API reference

QFT
QuadraticForm
QuantumVolume
RC3XGate
RCCXGate
RealAmplitudes
Reset
RGate
RGQFTMultiplier
RVGate
RXGate
RXXGate
RYGate

QFT

`qiskit.circuit.library.QFT(num_qubits=None, approximation_degree=0, do_swaps=True, inverse=False, insert_barriers=False, name=None)` GitHub ↗

Bases: `BlueprintCircuit`

Quantum Fourier Transform Circuit.

The Quantum Fourier Transform (QFT) on n qubits is the operation

$$|j\rangle \mapsto \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi ijk/2^n} |k\rangle$$

QFT implementation

QFT implementations on the bibliography

- Very detailed description: [Thomas C. Wong]
- QUIRK and QISKIT: QFT and IQFT are standard blocks.

IBM Quantum **Documentation** | Overview | Start | Build | Transpile | Verify | Run | API reference

- **QFT**
- QuadraticForm
- QuantumVolume
- RC3XGate
- RCCXGate
- RealAmplitudes
- Reset
- RGate
- RGQFTMultiplier
- RVGate
- RXGate
- RXXGate
- RYGate

QFT

`qiskit.circuit.library.QFT(num_qubits=None, approximation_degree=0, do_swaps=True, inverse=False, insert_barriers=False, name=None)` GitHub ↗

Bases: `BlueprintCircuit`

Quantum Fourier Transform Circuit.

The Quantum Fourier Transform (QFT) on n qubits is the operation

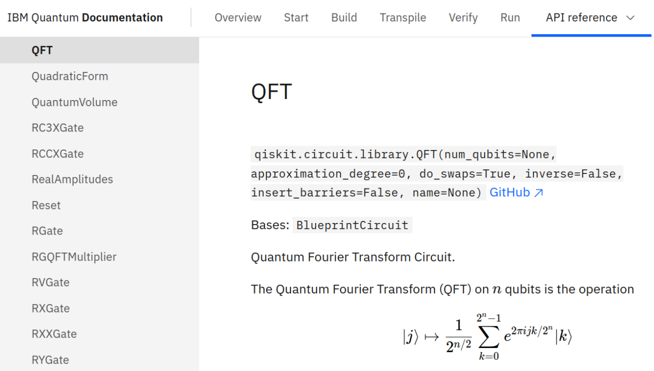
$$|j\rangle \mapsto \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi ijk/2^n} |k\rangle$$

- Custom QFT and IQFT, for academic purposes.

QFT implementation

QFT implementations on the bibliography

- Very detailed description: [Thomas C. Wong]
- QUIRK and QISKIT: QFT and IQFT are standard blocks.


IBM Quantum **Documentation** | Overview | Start | Build | Transpile | Verify | Run | API reference

QFT
QuadraticForm
QuantumVolume
RC3XGate
RCCXGate
RealAmplitudes
Reset
RGate
RGQFTMultiplier
RVGate
RXGate
RXXGate
RYGate

QFT

`qiskit.circuit.library.QFT(num_qubits=None, approximation_degree=0, do_swaps=True, inverse=False, insert_barriers=False, name=None)` GitHub ↗

Bases: `BlueprintCircuit`

Quantum Fourier Transform Circuit.

The Quantum Fourier Transform (QFT) on n qubits is the operation

$$|j\rangle \mapsto \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i jk/2^n} |k\rangle$$

- Custom QFT and IQFT, for academic purposes.
- Unless a good reason exists! Exotic optimization...

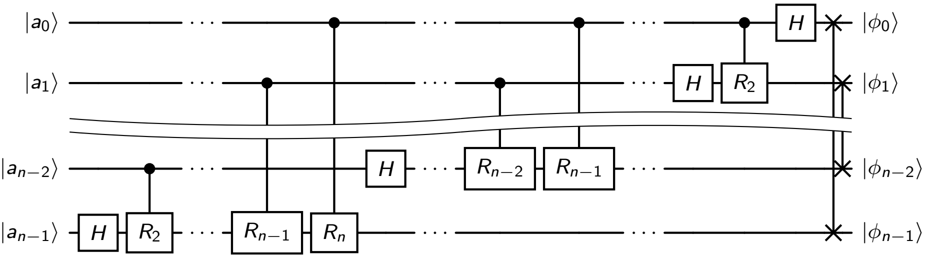
QFT implementation

[Thomas C. Wong]

If we define the R_k operator as:

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^r} \end{pmatrix},$$

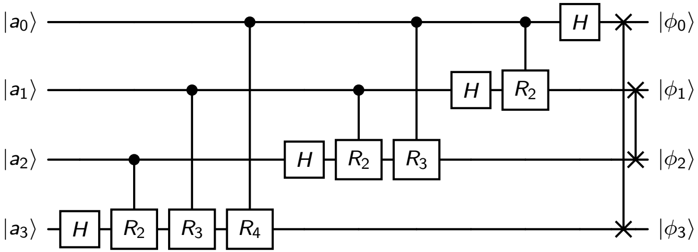
then the QFT operator looks like:



QFT implementation

[Thomas C. Wong], $N = 4$ qubits

Taking into account that $R^2 = Z^{1/2} = S$, $R_3 = Z^{1/4} = T$, $R_4 = Z^{1/8}$, the QFT operator for $N = 4$ qubits looks like:



LABORATORY EXERCISE 1

Implement on QUIRK the QFT for $N = 6$ qubits.
Use https://bit.ly/3SDNSxl from [Thomas C. Wong].

QFT implementation

LABORATORY EXERCISE 2

1. Check that QUIRK $\mathrm{QFT}^\dagger$ is able to reverse your QFT operator.
2. Implement on QUIRK the $\mathrm{QFT}^\dagger$
3. Check that your $\mathrm{QFT}^\dagger$ reverses QFT QUIRK operator.
4. Implement both QFT and $\mathrm{QFT}^\dagger$ and check that their combined action is the identity.

Final Question

What is the Quantum Fourier Transform of state $|0\rangle$? This would be equivalent to initializing to $|0\rangle$ all the qubits of the *n*-qubit register.

Outline

Phase Estimation - Algorithm, Step 1

Introduction to the Phase Estimation Algorithm

- We have a quantum gate U whose effect over a $|v\rangle$ state is a global phase θ.

Algorithm, Step 1

Phase Estimation - Algorithm, Step 1

Introduction to the Phase Estimation Algorithm

- We have a quantum gate U whose effect over a $|v\rangle$ state is a global phase θ.
- This is equivalent to $|v\rangle$ being an eigenstate of U with eigenvalue $e^{i\theta}$.

Algorithm, Step 1

Phase Estimation - Algorithm, Step 1

Introduction to the Phase Estimation Algorithm

- We have a quantum gate U whose effect over a $|v\rangle$ state is a global phase θ.
- This is equivalent to $|v\rangle$ being an eigenstate of U with eigenvalue $e^{i\theta}$.
- Let's consider gates U, U^2, U^4, U^8,..., $U^{2^{(n-1)}}$, where $n \in \mathbb{N}$ is the number of qubits of the registrer that is going to contain the phase estimation.

Algorithm, Step 1

Phase Estimation - Algorithm, Step 1

Introduction to the Phase Estimation Algorithm

- We have a quantum gate U whose effect over a $|v\rangle$ state is a global phase θ.
- This is equivalent to $|v\rangle$ being an eigenstate of U with eigenvalue $e^{i\theta}$.
- Let's consider gates U, U^2, U^4, U^8,..., $U^{2^{(n-1)}}$, where $n \in \mathbb{N}$ is the number of qubits of the registrer that is going to contain the phase estimation.
- Each time we apply a gate U^{2^l}, that state is multiplied by $e^{i\theta \cdot 2^l}$.

Algorithm, Step 1

Phase Estimation - Algorithm, Step 1

Introduction to the Phase Estimation Algorithm

- We have a quantum gate U whose effect over a $|v\rangle$ state is a global phase θ.
- This is equivalent to $|v\rangle$ being an eigenstate of U with eigenvalue $e^{i\theta}$.
- Let's consider gates U, U^2, U^4, U^8,..., $U^{2^{(n-1)}}$, where $n \in \mathbb{N}$ is the number of qubits of the registrer that is going to contain the phase estimation.
- Each time we apply a gate U^{2^l}, that state is multiplied by $e^{i\theta \cdot 2^l}$.

Algorithm, Step 1

1. Initialize a n-qubit register $(N = 2^n)$ with the state $|+\rangle \otimes |+\rangle \otimes \cdots \otimes + = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle$.

Phase Estimation - Algorithm, Step 1

Introduction to the Phase Estimation Algorithm

- We have a quantum gate U whose effect over a $|v\rangle$ state is a global phase θ.
- This is equivalent to $|v\rangle$ being an eigenstate of U with eigenvalue $e^{i\theta}$.
- Let's consider gates U, U^2, U^4, U^8,..., $U^{2^{(n-1)}}$, where $n \in \mathbb{N}$ is the number of qubits of the registrer that is going to contain the phase estimation.
- Each time we apply a gate U^{2^l}, that state is multiplied by $e^{i\theta \cdot 2^l}$.

Algorithm, Step 1

1. Initialize a n-qubit register $(N = 2^n)$ with the state $|+\rangle \otimes |+\rangle \otimes \cdots \otimes + = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle$.
2. Each gate U^{2^l}, controlled by qubit l of the n-qubit register.

Phase Estimation - Algorithm, Step 2

Phase Estimation - Step 2

The output of step 1 of the algorithm is

$$\mathrm{STEP}_1 \left| \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle \right) \otimes |v\rangle \right\rangle = \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{i\theta \cdot k} |k\rangle \right) \otimes |v\rangle$$

Phase Estimation - Algorithm, Step 2

Phase Estimation - Step 2

The output of step 1 of the algorithm is

$$\mathrm{STEP}_1 \left| \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle \right) \otimes |v\rangle \right\rangle = \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{i\theta \cdot k} |k\rangle \right) \otimes |v\rangle$$

Let's compare this expression with the QFT over $|a\rangle = \sum_{j=0}^{N-1} a_j |j\rangle$,

$$\mathrm{QFT}\, |a\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \sum_{j=0}^{N-1} a_j e^{2\pi i \cdot jk/N} |k\rangle$$

Phase Estimation - Algorithm, Step 2

Phase Estimation - Step 2

The output of step 1 of the algorithm is

$$\mathrm{STEP}_1 \left| \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle \right) \otimes |v\rangle \right\rangle = \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{i\theta \cdot k} |k\rangle \right) \otimes |v\rangle$$

Let's compare this expression with the QFT over $|a\rangle = \sum_{j=0}^{N-1} a_j |j\rangle$,

$$\mathrm{QFT}\,|a\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \sum_{j=0}^{N-1} a_j e^{2\pi i \cdot jk/N} |k\rangle$$

The output is actually $(\mathrm{QFT}\,|b\rangle) \otimes |v\rangle$, where

$$\theta = 2\pi b/N = 2\pi b/2^n = b \cdot (\pi/2^{n-1})$$

Phase Estimation - Algorithm, Step 2

Phase Estimation - Step 2

The output of step 1 of the algorithm is

$$\mathrm{STEP}_1 \left| \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle \right) \otimes |v\rangle \right\rangle = \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{i\theta \cdot k} |k\rangle \right) \otimes |v\rangle$$

Let's compare this expression with the QFT over $|a\rangle = \sum_{j=0}^{N-1} a_j |j\rangle$,

$$\mathrm{QFT} |a\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \sum_{j=0}^{N-1} a_j e^{2\pi i \cdot jk/N} |k\rangle$$

The output is actually $(\mathrm{QFT} |b\rangle) \otimes |v\rangle$, where

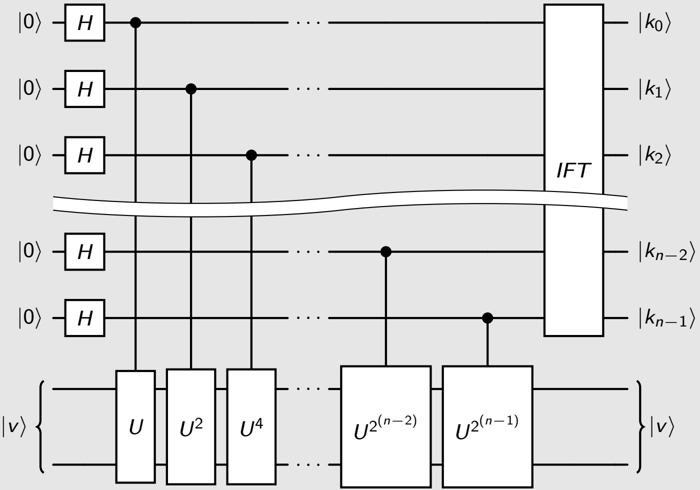
$$\theta = 2\pi b/N = 2\pi b/2^n = b \cdot (\pi/2^{n-1})$$

Last Sep

$\Rightarrow$ APPLY THE IQFT to obtain b $\Leftarrow$

Phase Estimation - Circuit

Phase Estimation - Circuit



Outline

Basic concepts

Basic properties

$$(ABC)^\dagger = C^\dagger B^\dagger A^\dagger, \qquad \langle \mathbf{v} | A\mathbf{w} \rangle = \langle A^\dagger \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{v} | A | \mathbf{w} \rangle$$

Basic concepts

Basic properties

$$(ABC)^\dagger = C^\dagger B^\dagger A^\dagger, \qquad \langle \mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger \mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|A|\mathbf{w}\rangle$$

Eigenvalues and Eigenvectors

$$A\mathbf{v} = \lambda \mathbf{v},$$

$\mathbf{v}$, eigenvector of A with associated eigenvalue λ.

Basic concepts

Basic properties

$$(ABC)^\dagger = C^\dagger B^\dagger A^\dagger, \qquad \langle \mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger \mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|A|\mathbf{w}\rangle$$

Eigenvalues and Eigenvectors

$$A\mathbf{v} = \lambda \mathbf{v},$$

$\mathbf{v}$, eigenvector of A with associated eigenvalue λ.

Hermitian (self-adjoint) Matrices

$$A = A^\dagger, \qquad \langle \mathbf{v}|A\mathbf{w}\rangle = \langle A\mathbf{v}|\mathbf{w}\rangle$$

$\mathbf{v}$, eigenvector of A with associated eigenvalue λ.

Basic concepts

Basic properties

$$(ABC)^\dagger = C^\dagger B^\dagger A^\dagger, \qquad \langle \mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger \mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|A|\mathbf{w}\rangle$$

Eigenvalues and Eigenvectors

$$A\mathbf{v} = \lambda \mathbf{v},$$

$\mathbf{v}$, eigenvector of A with associated eigenvalue λ.

Hermitian (self-adjoint) Matrices

$$A = A^\dagger, \qquad \langle \mathbf{v}|A\mathbf{w}\rangle = \langle A\mathbf{v}|\mathbf{w}\rangle$$

$\mathbf{v}$, eigenvector of A with associated eigenvalue λ.

Unitary Matrices

$$AA^\dagger = A^\dagger A = I, \qquad A^{-1} = A^\dagger, \qquad \langle A\mathbf{v}|A\mathbf{w}\rangle = \langle \mathbf{v}|\mathbf{w}\rangle$$

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle, \qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle$, $\qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u}|\mathbf{v}+\mathbf{w}\rangle = \langle \mathbf{u}|\mathbf{v}\rangle + \langle \mathbf{u}|\mathbf{w}\rangle$, $\qquad \langle \mathbf{u}+\mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{u}|\mathbf{w}\rangle + \langle \mathbf{v}|\mathbf{w}\rangle$
- $\langle \mathbf{u}|\alpha\mathbf{v}\rangle = \alpha \langle \mathbf{u}|\mathbf{v}\rangle$,
- $\langle \alpha\mathbf{u}|\mathbf{v}\rangle = \alpha^* \langle \mathbf{u}|\mathbf{v}\rangle$

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle$, $\qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,
- $\langle \alpha \mathbf{u} | \mathbf{v} \rangle = \alpha^* \langle \mathbf{u} | \mathbf{v} \rangle$
- $\overline{\langle \mathbf{u} | \mathbf{v} \rangle} = \langle \mathbf{u} | \mathbf{v} \rangle^* = \langle \mathbf{v} | \mathbf{u} \rangle$

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle$, $\qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,
- $\langle \alpha \mathbf{u} | \mathbf{v} \rangle = \alpha^* \langle \mathbf{u} | \mathbf{v} \rangle$
- $\overline{\langle \mathbf{u} | \mathbf{v} \rangle} = \langle \mathbf{u} | \mathbf{v} \rangle^* = \langle \mathbf{v} | \mathbf{u} \rangle$
- $\langle \mathbf{u} | \mathbf{u} \rangle \geq 0$, being $\langle \mathbf{u} | \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle$, $\qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,
- $\langle \alpha \mathbf{u} | \mathbf{v} \rangle = \alpha^* \langle \mathbf{u} | \mathbf{v} \rangle$
- $\overline{\langle \mathbf{u} | \mathbf{v} \rangle} = \langle \mathbf{u} | \mathbf{v} \rangle^* = \langle \mathbf{v} | \mathbf{u} \rangle$
- $\langle \mathbf{u} | \mathbf{u} \rangle \geq 0$, being $\langle \mathbf{u} | \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$
- Unitary base: orthonormal with the inner product $\langle \cdot | \cdot \rangle$

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle, \qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,
- $\langle \alpha \mathbf{u} | \mathbf{v} \rangle = \alpha^* \langle \mathbf{u} | \mathbf{v} \rangle$
- $\overline{\langle \mathbf{u} | \mathbf{v} \rangle} = \langle \mathbf{u} | \mathbf{v} \rangle^* = \langle \mathbf{v} | \mathbf{u} \rangle$
- $\langle \mathbf{u} | \mathbf{u} \rangle \geq 0$, being $\langle \mathbf{u} | \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$
- Unitary base: orthonormal with the inner product $\langle \cdot | \cdot \rangle$

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle$, $\qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,
- $\langle \alpha \mathbf{u} | \mathbf{v} \rangle = \alpha^* \langle \mathbf{u} | \mathbf{v} \rangle$
- $\overline{\langle \mathbf{u} | \mathbf{v} \rangle} = \langle \mathbf{u} | \mathbf{v} \rangle^* = \langle \mathbf{v} | \mathbf{u} \rangle$
- $\langle \mathbf{u} | \mathbf{u} \rangle \geq 0$, being $\langle \mathbf{u} | \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$
- Unitary base: orthonormal with the inner product $\langle \cdot | \cdot \rangle$

Hermitian (self-adjoint) Matrices

- $A = A^\dagger$, $a_{ij} = \bar{a}_{ji} = a_{ji}^*$

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle$, $\qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,
- $\langle \alpha \mathbf{u} | \mathbf{v} \rangle = \alpha^* \langle \mathbf{u} | \mathbf{v} \rangle$
- $\overline{\langle \mathbf{u} | \mathbf{v} \rangle} = \langle \mathbf{u} | \mathbf{v} \rangle^* = \langle \mathbf{v} | \mathbf{u} \rangle$
- $\langle \mathbf{u} | \mathbf{u} \rangle \geq 0$, being $\langle \mathbf{u} | \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$
- Unitary base: orthonormal with the inner product $\langle \cdot | \cdot \rangle$

Hermitian (self-adjoint) Matrices

- $A = A^\dagger$, $a_{ij} = \bar{a}_{ji} = a_{ji}^*$
- $\langle \mathbf{v} | A \mathbf{w} \rangle = \langle A^\dagger \mathbf{v} | \mathbf{w} \rangle = \langle A \mathbf{v} | \mathbf{w} \rangle$

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle$, $\quad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,
- $\langle \alpha \mathbf{u} | \mathbf{v} \rangle = \alpha^* \langle \mathbf{u} | \mathbf{v} \rangle$
- $\overline{\langle \mathbf{u} | \mathbf{v} \rangle} = \langle \mathbf{u} | \mathbf{v} \rangle^* = \langle \mathbf{v} | \mathbf{u} \rangle$
- $\langle \mathbf{u} | \mathbf{u} \rangle \geq 0$, being $\langle \mathbf{u} | \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$
- Unitary base: orthonormal with the inner product $\langle \cdot | \cdot \rangle$

Hermitian (self-adjoint) Matrices

- $A = A^\dagger$, $a_{ij} = \bar{a}_{ji} = a_{ji}^*$
- $\langle \mathbf{v} | A\mathbf{w} \rangle = \langle A^\dagger \mathbf{v} | \mathbf{w} \rangle = \langle A\mathbf{v} | \mathbf{w} \rangle$
- A, diagonalizable with **real eigenvalues**

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle$, $\qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,
- $\langle \alpha \mathbf{u} | \mathbf{v} \rangle = \alpha^* \langle \mathbf{u} | \mathbf{v} \rangle$
- $\overline{\langle \mathbf{u} | \mathbf{v} \rangle} = \langle \mathbf{u} | \mathbf{v} \rangle^* = \langle \mathbf{v} | \mathbf{u} \rangle$
- $\langle \mathbf{u} | \mathbf{u} \rangle \geq 0$, being $\langle \mathbf{u} | \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$
- Unitary base: orthonormal with the inner product $\langle \cdot | \cdot \rangle$

Hermitian (self-adjoint) Matrices

- $A = A^\dagger$, $a_{ij} = \bar{a}_{ji} = a_{ji}^*$
- $\langle \mathbf{v} | A\mathbf{w} \rangle = \langle A^\dagger \mathbf{v} | \mathbf{w} \rangle = \langle A\mathbf{v} | \mathbf{w} \rangle$
- A, diagonalizable with **real eigenvalues**
- A has a unitary base of eigenvectors, eigenvectors from different eigenspaces are orthogonal

Hermitian Inner Product & Hermitian (self-adjoint) Matrices

Hermitian Inner Product: relation with scalar product

- $\langle \mathbf{u} | \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{v} \rangle + \langle \mathbf{u} | \mathbf{w} \rangle$, $\qquad \langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
- $\langle \mathbf{u} | \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u} | \mathbf{v} \rangle$,
- $\langle \alpha \mathbf{u} | \mathbf{v} \rangle = \alpha^* \langle \mathbf{u} | \mathbf{v} \rangle$
- $\overline{\langle \mathbf{u} | \mathbf{v} \rangle} = \langle \mathbf{u} | \mathbf{v} \rangle^* = \langle \mathbf{v} | \mathbf{u} \rangle$
- $\langle \mathbf{u} | \mathbf{u} \rangle \geq 0$, being $\langle \mathbf{u} | \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$
- Unitary base: orthonormal with the inner product $\langle \cdot | \cdot \rangle$

Hermitian (self-adjoint) Matrices

- $A = A^\dagger$, $a_{ij} = \bar{a}_{ji} = a_{ji}^*$
- $\langle \mathbf{v} | A\mathbf{w} \rangle = \langle A^\dagger \mathbf{v} | \mathbf{w} \rangle = \langle A\mathbf{v} | \mathbf{w} \rangle$
- A, diagonalizable with **real eigenvalues**
- A has a unitary base of eigenvectors, eigenvectors from different eigenspaces are orthogonal
- If A is real, *symmetric matrix*

Unitary Matrices

Unitary Matrices

- Quantum gates: unitary operators

Unitary Matrices

Unitary Matrices

- Quantum gates: unitary operators
- $AA^\dagger = A^\dagger A = I$, $A^{-1} = A^\dagger$

Unitary Matrices

Unitary Matrices

- Quantum gates: unitary operators
- $AA^\dagger = A^\dagger A = I$, $A^{-1} = A^\dagger$
- $\langle A\mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger A\mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|\mathbf{w}\rangle$

Unitary Matrices

Unitary Matrices

- Quantum gates: unitary operators
- $AA^\dagger = A^\dagger A = I$, $A^{-1} = A^\dagger$
- $\langle A\mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger A\mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|\mathbf{w}\rangle$
- Columns and rows of A, **unitary bases**

Unitary Matrices

Unitary Matrices

- Quantum gates: unitary operators
- $AA^\dagger = A^\dagger A = I$, $A^{-1} = A^\dagger$
- $\langle A\mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger A\mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|\mathbf{w}\rangle$
- Columns and rows of A, **unitary bases**
- A, **diagonalizable**

Unitary Matrices

Unitary Matrices

- Quantum gates: unitary operators
- $AA^\dagger = A^\dagger A = I$, $A^{-1} = A^\dagger$
- $\langle A\mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger A\mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|\mathbf{w}\rangle$
- Columns and rows of A, **unitary bases**
- A, **diagonalizable**
- If α is an eigenvalue of A, $|\alpha| = 1$

Unitary Matrices

Unitary Matrices

- Quantum gates: unitary operators
- $AA^\dagger = A^\dagger A = I$, $A^{-1} = A^\dagger$
- $\langle A\mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger A\mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|\mathbf{w}\rangle$
- Columns and rows of A, **unitary bases**
- A, **diagonalizable**
- If α is an eigenvalue of A, $|\alpha| = 1$
- A has a unitary base of eigenvectors, eigenvectors from different eigenspaces are orthogonal

Unitary Matrices

Unitary Matrices

- Quantum gates: unitary operators
- $AA^\dagger = A^\dagger A = I$, $A^{-1} = A^\dagger$
- $\langle A\mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger A\mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|\mathbf{w}\rangle$
- Columns and rows of A, **unitary bases**
- A, **diagonalizable**
- If α is an eigenvalue of A, $|\alpha| = 1$
- A has a unitary base of eigenvectors, eigenvectors from different eigenspaces are orthogonal
- $A = e^{iH}$ for some H Hermitian

Unitary Matrices

Unitary Matrices

- Quantum gates: unitary operators
- $AA^\dagger = A^\dagger A = I$, $A^{-1} = A^\dagger$
- $\langle A\mathbf{v}|A\mathbf{w}\rangle = \langle A^\dagger A\mathbf{v}|\mathbf{w}\rangle = \langle \mathbf{v}|\mathbf{w}\rangle$
- Columns and rows of A, **unitary bases**
- A, **diagonalizable**
- If α is an eigenvalue of A, $|\alpha| = 1$
- A has a unitary base of eigenvectors, eigenvectors from different eigenspaces are orthogonal
- $A = e^{iH}$ for some H Hermitian
- $e^{iH} = \sum_{k=0}^{\infty} \frac{1}{k!}(iH)^k = I + iH - \frac{1}{2}H^2 - \frac{i}{6}H^3 + \frac{1}{24}H^4 + \ldots$

Laboratory Exercises (Matlab / Octave) (1/3)

Hermitian Inner Product

1. Check that Matlab computes the correct values (0 in all the cases) for these expressions
   - $\langle \mathbf{u}+\mathbf{v}|\mathbf{w}\rangle - \langle \mathbf{u}|\mathbf{w}\rangle - \langle \mathbf{v}|\mathbf{w}\rangle$
   - $\langle \mathbf{u}|\mathbf{v}+\mathbf{w}\rangle - \langle \mathbf{u}|\mathbf{v}\rangle - \langle \mathbf{u}|\mathbf{w}\rangle$
   - $\langle \mathbf{u}|a\mathbf{v}\rangle - a\,\langle \mathbf{u}|\mathbf{v}\rangle$
   - $\langle a\mathbf{u}|\mathbf{v}\rangle - a^*\,\langle \mathbf{u}|\mathbf{v}\rangle$
   - $\langle \mathbf{u}|\mathbf{v}\rangle - \overline{\langle \mathbf{v}|\mathbf{u}\rangle}$

   In all the cases, $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w} \in \mathbb{C}^4$, $a \in \mathbb{C}$. The entrances are random numbers generated via `rand(N)+i*rand(N)`
2. Compute the values of $|\mathbf{u}|^2 = \langle \mathbf{u}|\mathbf{u}\rangle$, $|\mathbf{v}|^2$ and $|\mathbf{w}|^2$.

Laboratory Exercises (Matlab / Octave) (2/3)

Generation of matrices U (Unitary) and H (Hermitian)

1. Random matrix $A \in \mathcal{M}_{N\times N}(\mathbb{C})$ with $N = 8$:

   `N=8;A=rand(N,N)+i*rand(N,N);`
2. Generate an Hermitian matrix via $H = A + A^{\dagger}$
3. Generate a Unitary matrix U from A: `[U,~]=qr(A)` Note that the QR decomposition of any square matrix A is: $A = QR$, where Q is unitary and F is upper triangular.

Exercises

Use `norm(A)` *to verify that A is machine-precision nul. I.e., check whether $U \cdot U' = I$ via* `norm(U*U'-eye(N))`

1. Check that A is neither Hermitian nor Unitary
2. Check that H is Hermitian and not Unitary
3. Check that U is Unitary and not Hermitian
4. Check that $V = e^{iH}$ (via `expm`) is Unitary and not Hermitian

Laboratory Exercises (Matlab / Octave) (3/3)

Exercises: eigenvectors and eigenvalues

By using `[V,lambda] = eig(A)` for computing the base of eigenvalues (V) and the diagonal matrix (λ), answer these questions:

1. Compute the eigenvectors, the eigenvalues and the absolute values of the eigenvalues of A
2. The same for H
3. The same for U
4. Check that the bases of eigenvectors of H and U are unitary

Outline

Outline

Multiple Eigenvalues

Multiple Eigenvalues

1. Since U is unitary, all its eigenvalues have unitary module.

Multiple Eigenvalues

Multiple Eigenvalues

1. Since U is unitary, all its eigenvalues have unitary module.
2. Idea: write the initial state $|v\rangle$ on the Phase Estimation Algorithm on a eigenvector basis.

Multiple Eigenvalues

Multiple Eigenvalues

1. Since U is unitary, all its eigenvalues have unitary module.
2. Idea: write the initial state $|v\rangle$ on the Phase Estimation Algorithm on a eigenvector basis.
3. Since the Phase Estimation Algorithm is linear, we should recover a linear composition of states. These are, precisely, the eigenvalues.

Multiple Eigenvalues

Multiple Eigenvalues

1. Since U is unitary, all its eigenvalues have unitary module.
2. Idea: write the initial state $|v\rangle$ on the Phase Estimation Algorithm on a eigenvector basis.
3. Since the Phase Estimation Algorithm is linear, we should recover a linear composition of states. These are, precisely, the eigenvalues.
4. Each time we run the Phase Estimation Algorithm, one of the eigenvalues is recovered.

Multiple Eigenvalues

Multiple Eigenvalues

1. Since U is unitary, all its eigenvalues have unitary module.
2. Idea: write the initial state $|v\rangle$ on the Phase Estimation Algorithm on a eigenvector basis.
3. Since the Phase Estimation Algorithm is linear, we should recover a linear composition of states. These are, precisely, the eigenvalues.
4. Each time we run the Phase Estimation Algorithm, one of the eigenvalues is recovered.
5. The probability of recovering a particular eigenvalue will depend on the squared amplitude of the corresponding eigenvector on the initial state $|v\rangle$.

Laboratory Exercises on QUIRK, Multiple Eigenvalues

General instructions

- In all the cases, you should adjoin the QUIRK quantum circuit.
- The code `octave.m` can be found on Moodle.
- If you have to use `octave.m` code to generate a matrix, please adjoin the corresponding files `out.txt` and `matrix.txt` for each case.

QUIRK exercises

1. Use $N = 2$ and 3 qubits for $|b\rangle$ (quantum register containing the phase estimation). Gate U (whose phase is being estimate) will be Y and $Y^{1/2}$
2. Use $N = 3$ and 4 qubits for $|b\rangle$; for U use:
   - a 1-qubit U quantum gate generated by `octave.m`
   - a 2-qubit U quantum gate generated by `octave.m`

Outline

Addition

Addition (1/2)

1. The phase estimation algorithm can be used for addition.

Addition

Addition (1/2)

1. The phase estimation algorithm can be used for addition.
2. Based on the very well-known equation $e^{a+b} = e^a \cdot e^b$: the composition of phase gates is the sum of the phases.

Addition

Addition (1/2)

1. The phase estimation algorithm can be used for addition.
2. Based on the very well-known equation $e^{a+b} = e^a \cdot e^b$: the composition of phase gates is the sum of the phases.
3. Hence, we can sum powers of 2, 2^k, to a n-qubit register, by *estimating* the phase of a gate acting on a eigenvector with eigenvalue $e^{i\pi \cdot k/2^{n-1}}$.

Addition

Addition (1/2)

1. The phase estimation algorithm can be used for addition.
2. Based on the very well-known equation $e^{a+b} = e^a \cdot e^b$: the composition of phase gates is the sum of the phases.
3. Hence, we can sum powers of 2, 2^k, to a n-qubit register, by *estimating* the phase of a gate acting on a eigenvector with eigenvalue $e^{i\pi \cdot k/2^{n-1}}$.
4. The sum is decomposed as $a + k = a + \sum_{l=0}^{n-1} 2^l \cdot k_l$.

Addition

Addition (2/2)

1. We use the qubits of the register $|k\rangle$ to control the phase gates we are meausering.

Addition

Addition (2/2)

1. We use the qubits of the register $|k\rangle$ to control the phase gates we are meausering.
2. Even better, we just apply fractions of the Z gate on the $|k\rangle$ register qubits controlled by the n-qubit register that is undergoing the IQFT.

Addition

Addition (2/2)

1. We use the qubits of the register $|k\rangle$ to control the phase gates we are meausering.
2. Even better, we just apply fractions of the Z gate on the $|k\rangle$ register qubits controlled by the n-qubit register that is undergoing the IQFT.
3. $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle = e^{i\pi}|1\rangle$, $Z^{1/m}|1\rangle = e^{i\pi/m}|1\rangle$

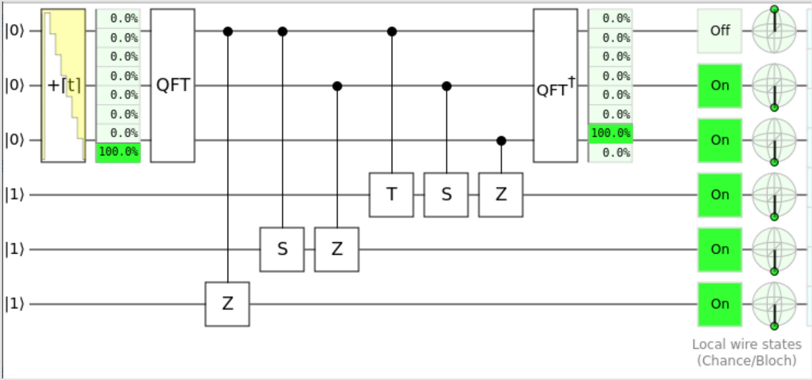
Addition

Addition (2/2)

1. We use the qubits of the register $|k\rangle$ to control the phase gates we are meausering.
2. Even better, we just apply fractions of the Z gate on the $|k\rangle$ register qubits controlled by the n-qubit register that is undergoing the IQFT.
3. $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle = e^{i\pi}|1\rangle$, $Z^{1/m}|1\rangle = e^{i\pi/m}|1\rangle$
4. One of the addings $+a$ can be applying the QFT over the n-qubit register initialized to $|a\rangle$ instead of initializing all the n-qubit register with Hadamard gates $(\mathrm{QFT}|0\rangle)$.

Example

Example (3-qubit on QUIRK)



Laboratory Exercise

Exercise

1. Implement on QUIRK the quantum addition of two 4-qubit numbers $|a\rangle$ and $|b\rangle$. The input is a 8-qubit register, $|a\rangle \times |b\rangle$.

Laboratory Exercise

Exercise

1. Implement on QUIRK the quantum addition of two 4-qubit numbers $|a\rangle$ and $|b\rangle$. The input is a 8-qubit register, $|a\rangle \times |b\rangle$.
2. Sum $+\lceil t \rceil$ and $|3\rangle$ counters.

Laboratory Exercise

Exercise

1. Implement on QUIRK the quantum addition of two 4-qubit numbers $|a\rangle$ and $|b\rangle$. The input is a 8-qubit register, $|a\rangle \times |b\rangle$.
2. Sum $+\lceil t \rceil$ and $|3\rangle$ counters.
3. Sum $+\lceil t \rceil$ and $-\lceil t \rceil$ counters.

Laboratory Exercise

Exercise

1. Implement on QUIRK the quantum addition of two 4-qubit numbers $|a\rangle$ and $|b\rangle$. The input is a 8-qubit register, $|a\rangle \times |b\rangle$.
2. Sum $+\lceil t \rceil$ and $|3\rangle$ counters.
3. Sum $+\lceil t \rceil$ and $-\lceil t \rceil$ counters.
4. The input is on an entangled state $\frac{1}{\sqrt{2}}(|2\rangle \times |9\rangle + |6\rangle \times |5\rangle)$.

Laboratory Exercise

Exercise

1. Implement on QUIRK the quantum addition of two 4-qubit numbers $|a\rangle$ and $|b\rangle$. The input is a 8-qubit register, $|a\rangle \times |b\rangle$.
2. Sum $+\lceil t \rceil$ and $|3\rangle$ counters.
3. Sum $+\lceil t \rceil$ and $-\lceil t \rceil$ counters.
4. The input is on an entangled state $\frac{1}{\sqrt{2}}(|2\rangle \times |9\rangle + |6\rangle \times |5\rangle)$.
5. Did you expected these results? Please, discuss.

Outline

Shor's Algorithm

Shor's Algorithm

- This is NOT a usual Quantum Search Algorithm. This is neither the Grover Algorithm nor any kind of Quantum Walk Algorithm.

Shor's Algorithm

Shor's Algorithm

- This is NOT a usual Quantum Search Algorithm. This is neither the Grover Algorithm nor any kind of Quantum Walk Algorithm.
- Instead, Shor's Algorithm translates the factorization problem into a Eigenvalue problem.

Shor's Algorithm

Shor's Algorithm

- This is NOT a usual Quantum Search Algorithm. This is neither the Grover Algorithm nor any kind of Quantum Walk Algorithm.
- Instead, Shor's Algorithm translates the factorization problem into a Eigenvalue problem.
- This Eigenvalue problem can be solved with the Phase Estimation Algorithm.

Shor's Algorithm

Shor's Algorithm

- This is NOT a usual Quantum Search Algorithm. This is neither the Grover Algorithm nor any kind of Quantum Walk Algorithm.
- Instead, Shor's Algorithm translates the factorization problem into a Eigenvalue problem.
- This Eigenvalue problem can be solved with the Phase Estimation Algorithm.
- Fundaments of Shor's Algorithm: the Quantum Fourier Transform.

References

- Thomas C. Wong. Introduction to Classical and Quantum Computing. https://www.thomaswong.net
- N. D. Mermin. Quantum Computer Science, Cambrigde University Press.
- M.A. Nielsen and I.L. Chuang. Quantum Computation and Quantum Information, Cambrigde University Press.
- IBM Quantum Learning: https://learning.quantum.ibm.com
- IBM Quantum Documentation, QFT implementation: https://docs.quantum.ibm.com/api/qiskit/qiskit.circuit.library.QFT